

Common **interview** questions for **ML Engineers**



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1. ML fundamentals



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- 1 What is overfitting, and how can you prevent it?
 - 2 How does the k-nearest neighbors (KNN) algorithm work?
 - 3 What is the difference between L1 and L2 regularization?
 - 4 Explain the concept of feature scaling and why it's important.
 - 5 What is cross-validation, and why is it used?



2. Deep learning & Neural networks



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- 1** What is batch normalization, and how does it help in training deep networks?
 - 2** What is an attention mechanism, and why is it important in deep learning models?
 - 3** Explain the concept of transfer learning.
 - 4** What is the purpose of embedding layers in neural networks?
 - 5** What is the difference between CNNs and RNNs?



3. ML System design



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- 1** How would you design a recommendation system for a streaming service?
 - 2** What considerations are important when designing a real-time bidding system for online advertising?
 - 3** How would you architect a scalable ML pipeline for processing large datasets?
 - 4** What strategies would you employ to ensure data privacy in the ML system?
 - 5** How do you handle model versioning and deployment in a CI/CD environment?



4. Coding & data structures



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- 1** Implement k-nearest neighbors from scratch in Python.
 - 2** Write a function to detect a cycle in a linked list.
 - 3** Write code to perform a depth-first search (DFS) on a graph.
 - 4** Implement a binary search algorithm.
 - 5** Write a function to find the longest common prefix among an array of strings.



5. Probability & Statistics



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- 1** What is the difference between covariance and correlation?
 - 2** Explain the concept of a p-value in hypothesis testing.
 - 3** Can you explain Bayes Theorem in simple terms?
 - 4** How do you interpret a confidence interval?
 - 5** What is the law of large numbers, and why is it important?



6. Problem-solving & case studies



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- 1** How would you detect anomalies in a time-series dataset?
 - 2** Design a system to predict the likelihood of a user clicking on an online ad.
 - 3** How would you approach building a model to forecast stock prices?
 - 4** Develop a strategy to reduce false positives in a spam detection system.
 - 5** How would you evaluate the effectiveness of a ML model used for medical diagnosis?



7. Soft skills & communication



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- 1** Explain a complex ML concept to a non-technical stakeholder.
 - 2** How do you handle disagreements within a team regarding the approach to a project?
 - 3** How do you prioritize multiple ML projects with tight deadlines?
 - 4** How do you ensure effective communication when working remotely with team members?
 - 5** Can you provide an example of how you handled feedback or criticism about your work from a colleague or supervisor?



8. Scenario-based questions

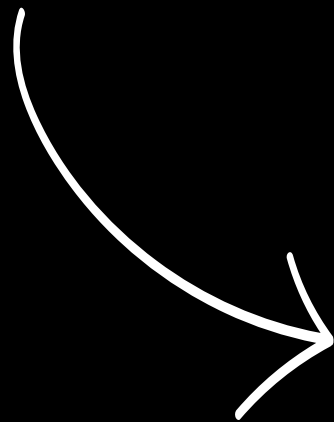


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- 1** Imagine you have a model that performs exceptionally well on training data but poorly on test data. What would you do?
 - 2** Suppose you're given a dataset with a large number of features. How would you determine which features to include?
 - 3** If you discover that your model is biased against a particular group, what actions would you take to handle this bias?
 - 4** Your model's predictions are accurate but not interpretable. How would you explain the results to stakeholders?
 - 5** Your team is behind schedule on delivering a machine learning project. What actions would you take to get the project back on track?



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